

Mathematical methods of managing a retail store as a dynamic system: the promotional pricing aspect

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The purpose of this study is to develop a framework that can address issues arising while managing a retail store as a complex system with stochastic elements. Specifically, we will focus on promotional pricing, which is a temporary reduction in price aimed at changing customer behavior. Promotional pricing is reflected in the category management paradigm, where each category is treated as a separate business entity. However, customers utility maximization strategy has a form:

$$P(f(\cdot), A) = \arg \max_{y \in A} f(y), \quad [1]$$

where A is a set of different categories.

So managing category independently much likely does not satisfy to the optimal control:

$$\Phi(u) \longrightarrow \max_{y \in A} f(y) \quad [1]$$

This problem of managing the disbalance between category-level and store-level objectives can be formally defined as a stochastic setting where the Center does not control agent actions directly, so the Center chooses control actions to respond to agents actions in multi-stage game in order to optimize its objectives.

$$\max_{u_t \in \mathcal{U}} \mathbb{E}_{a_t \sim \mathcal{P}_t} [R_{\text{global}}(a_t, u_{t-1}) + \gamma \cdot V_{t+1}(s_{t+1})]$$

where:

a_t - agents' (categories) actions at time t , drawn from $\mathcal{P}_t$,

u_{t-1} - Center's response chosen at previous step $t - 1$,

$R_{\text{global}}(a_t, u_{t-1})$ - revenue at time t , depending on a_t and prior Center response,

s_{t+1} - system state after actions (a_t, u_t) ,

$V_{t+1}(s_{t+1})$ - expected future revenue starting from next state,

γ - discount factor (optional).

The proposed solution combines game-theoretical formulations and optimization problems.

References

- [1] Novikov D. A. Teoriya upravleniya organizatsionnymi sistemami: vvodnyy kurs. - Moscow: IPU RAN, 2004. - Vol. 79. - P. 26.